

The impact of Strength of Schedule Balance on Tournament Efficacy

João P.L. Marchetti, João P.F. Batista, and Pedro O.S. Vaz-de-Melo

Computer Science Department, Universidade Federal de Minas Gerais (UFMG)

Abstract. Professional sports leagues rely on the double round-robin format to promote competitive fairness, yet the temporal ordering of opponents—whether a team faces a sequence of elite rivals early or begins with a more favorable schedule—remains a persistent source of debate. Theoretical accounts suggest that such sequencing may generate psychological feedback loops: front-loaded difficulty could induce a discouragement effect, whereas early success may foster momentum, potentially affecting final standings. This paper introduces the *Strength of Schedule Balance (SSB)* metric to quantify the temporal distribution of opponent difficulty and applies a stratified non-parametric framework to a validated dataset of 306 seasons from 20 professional football leagues ($N = 5,944$ schedules). Our results indicate that the double round-robin format is generally robust to schedule imbalance at the aggregate level, with most competitions showing no large or systematic distortions in final ranking outcomes attributable solely to fixture sequencing. However, league-level analysis reveals localized anomalies, particularly for unbalanced-weak schedules, where easier starts are more consistently associated with advantageous outcomes than harder starts are with disadvantageous ones. These findings suggest that schedule balance is not a universal determinant of tournament efficacy, but neither is it a purely logistical artifact; its impact appears to depend on the competitive environment of each league.

Keywords: Tournament Scheduling · Competitive Balance · Sports Analytics · Strength of Schedule · Causal Inference

1 Introduction

Professional sports leagues face a fundamental design challenge: ensuring that the tournament format yields a final ranking that accurately reflects the true skill levels of the participants. While several works [28, 21, 18, 37] emphasized the importance of outcome uncertainty for sustaining fan interest, the structural integrity of the schedule itself plays a distinct, yet equally critical, role in competitive fairness. As noted by Szymanski [33], the specific rules and organization of a contest shape the incentives for effort. As the sequence of opponents defines the competitive path a team must navigate, it can influence whether the final standings accurately reflect teams’ true latent abilities.

Traditionally, tournament scheduling has been treated primarily as an operations research problem focused on minimizing travel costs and logistical constraints [14, 11]. However, there is an ongoing discussion regarding whether scheduling significantly impacts the accuracy of team rankings. Empirical evidence suggests that the sequence

of matches matters. Krumer and Lechner [19], for instance, found that the ordering of opponents in round-robin tournaments can create significant “first-mover” advantages in knockout UEFA championships. In contrast, Goossens and Spieksma [15] found that “carry-over effects” [29]—where a team’s performance is influenced by the strength of its previous opponent—were non-existent in the Belgian football league.

Despite these insights, the consequences of *structural* schedule imbalance remain understudied. Contest theory suggests that “front-loaded” or “back-loaded” sequences can distort incentives, triggering either a “discouragement effect” [33]—where early losses reduce rational effort—or a “momentum effect” [9], where early success catalyzes performance. Consequently, it remains unclear whether such imbalances introduce systematic bias into final standings, thereby compromising the tournament’s efficacy in recovering the true ordering of team abilities.

This paper addresses this gap by introducing the *Strength of Schedule Balance* (SSB) to quantify the temporal distribution of opponent difficulty. Analyzing 20 major professional football leagues over 19 seasons [34], we compute the metric using two proxies for team strength—final league standing and team market value—and classify schedules as (i) **balanced**, (ii) **unbalanced-strong**, or (iii) **unbalanced-weak**. We then statistically assess whether these structural imbalances are associated with distortions in tournament efficacy, that is, with the league’s ability to recover latent team rankings.

Contrary to theoretical concerns regarding schedule fairness, our findings indicate that the standard double round-robin format is generally robust. While the aggregate evidence suggests that, in most competitions, intrinsic team quality outweighs scheduling nuances, the league-level analysis reveals localized anomalies. In particular, easier starts are more consistently associated with advantageous outcomes than harder starts are with disadvantageous ones, suggesting that the impact of schedule imbalance is conditional rather than universal. These findings challenge the view of fixture ordering as a purely logistical artifact and instead position schedule balance as a context-dependent factor in competitive fairness.

The remainder of this paper proceeds as follows. We first review relevant literature on tournament scheduling and efficacy determinants. We then present our **SSB** metric and describe the dataset. Subsequently, we report our empirical findings on both research questions before concluding with implications for league design.

2 Related Work

Tournament Scheduling and Fairness. Tournament scheduling has traditionally focused on operational constraints such as travel distance and fixture breaks, aiming to reduce costs for clubs and organizers [14]. However, current research recognizes that the schedule structure itself affects fairness and competitive balance by systematically favouring some teams through the timing of opponents [11, 36]. Bartsch, Drexl, and Kröger [3] introduced the partitioning of teams into strength groups to avoid long runs of matches against similar-quality opponents, a notion formalized by Briskorn and Knust [6] to mitigate carry-over effects, that is, situations where a team’s performance influences subsequent matches [29]. While designing “fair” schedules requires solving complex

combinatorial optimization problems [7], empirical evidence on how these operational imbalances translate to on-field outcomes remains scarce.

Tournament Efficacy and Competitive Incentives. Tournament efficacy refers to the ability of a format to recover the true latent ordering of team abilities. It is typically assessed by comparing the final ranking to an underlying “true” ranking of team strengths, using metrics such as Kendall’s τ , Spearman’s footrule distance, or the fraction of tournaments where the best team wins on Monte Carlo simulations [2, 20].

Several factors may impact tournament efficacy. The most extensively studied factor in the literature is tournament format [2, 32, 31, 22, 20, 10, 30] — with the double round-robin generally considered one of the most reliable formats — and seeding [12, 16, 17, 26], as noted by [11]. However, other factors have been shown to influence game outcomes which, if not regulated, can lead to diminished efficacy in the overall tournament, such as home advantage [24, 23] and referee bias [4].

While standard efficacy models treat match outcomes as stochastic processes, theoretical and empirical work suggests that schedule structure creates path dependence. Szymanski [33] argues that a sequence of elite opponents can trigger a “discouragement effect,” where early losses induce a rational reduction in effort. Conversely, favorable schedules may generate a “momentum effect”; Cohen-Zada, Krumer, and Shtudiner [9] demonstrate that past success causally increases the probability of future victory through psychological mechanisms, independent of strategic incentives. Consequently, if structural imbalances systematically induce either discouragement or momentum, the tournament’s efficacy in recovering the true ordering of team abilities is compromised.

Research Gap. Despite the extensive literature on constructing balanced schedules from an operations research perspective [5, 7, 8, 29, 35, 25], a significant disconnect remains between algorithmic scheduling and its actual impact on tournament efficacy. Existing empirical studies on schedule-induced bias often yield mixed conclusions and are typically limited to single competitions or specific knockout contexts [19, 15, 1, 13]. Consequently, there is a lack of a unified, quantitative framework to assess how the temporal distribution of opponent strength affects the accuracy of league standings. Specifically, the literature has yet to determine whether theoretical phenomena like the “discouragement effect” [33] or its counterpart, the “momentum effect” [9], are merely localized anomalies or systematic features of professional league structures.

3 Data

3.1 Data Acquisition and Source

The empirical analysis utilizes a longitudinal dataset of professional football competitions, restricted to the **2004–2024** period to ensure the synchronization of performance metrics with financial valuation data. Data were programmatically extracted via web scraping from *Transfermarkt* [34], a comprehensive repository for association football statistics. To facilitate a multi-granular analysis, the extraction pipeline targeted three distinct data structures:

1. **Final League Standings:** These records capture the aggregate seasonal performance for each team. Specifically, the dataset includes the final position, team name, total matches played, and the specific distribution of outcomes (won, drawn, and lost). Furthermore, we extracted the total points accumulated by each team, which are used to assess the efficacy of the league’s final ranking.
2. **Match-Level Records:** To reconstruct the chronological sequence of the season, we extracted granular fixture data. This includes temporal metadata — date and time — alongside the specific round (matchday) number. For each fixture, we identified the home and away teams, along with the official match result.
3. **Market Valuations:** We extracted the total market value of each participating squad, denominated in Euros (€). These values serve as a high-fidelity proxy for latent team strength, allowing the model to distinguish between schedule-induced performance variance and intrinsic quality disparities.

3.2 Data Preprocessing and Validation Protocols

To ensure the structural integrity of the analysis, a multi-stage preprocessing pipeline was implemented. Data were subjected to rigorous validation based on the following **business rules** to filter out anomalous or incomplete seasons:

- **Tournament Structure:** Inclusion was strictly limited to double round-robin formats. Competitions utilizing multi-stage playoffs or irregular structures that deviate from the standard home-and-away parity were excluded to maintain the consistency of our proposed Strength of Schedule Balance metric.
- **Operational Stability and Reporting Fidelity:** Seasons were retained only if at least 95% of match records contained non-null attendance values. While attendance is not a primary variable of interest, this threshold serves as a proxy for operational continuity and data reliability. The absence of attendance records often correlates with external disruptions—such as the COVID-19 pandemic, geopolitical instability, or administrative failures—which result in "behind-closed-doors" matches or irregular competitive conditions. By applying this filter, we ensure the dataset represents standardized seasons where environmental and institutional factors remained consistent.
- **Financial Data Completeness:** A season was retained only if market valuation data were available for all participating clubs. This ensures that every scheduling data point is paired with a corresponding financial proxy for team ability.

3.3 Dataset Composition

The final validated dataset comprises 306 seasons across 20 distinct leagues, representing 11 countries and 3 continents (Europe, Asia, and South America). Table 1 provides a detailed summary of the coverage and valid seasons per competition. The dataset exhibits a mean of 15.30 valid seasons per league, with the French *Ligue 1* providing one of the highest longitudinal depth (19 seasons). Conversely, the Turkish *Süper Lig* represents the most constrained subset due to stricter validation outcomes.

Table 1. Descriptive statistics of the validated longitudinal dataset by league. The *Avg. Teams* column reflects the mean number of participants per season, accounting for historical expansions or contractions in league size. *Coverage (%)* represents the data density, defined as the ratio of successfully validated seasons to the total chronological span from the *Start* year through 2024.

League Name	Seasons	Avg. Teams	Country	Start	Coverage (%)
Ligue 1	19	19.79	France	2004	90.48
Premier League	19	20.00	England	2004	90.48
Championship	19	24.00	England	2004	90.48
Serie A	19	20.00	Italy	2004	90.48
Serie B	19	21.42	Italy	2004	90.48
Liga Portugal	19	16.95	Portugal	2004	90.48
Bundesliga	18	18.00	Germany	2004	85.71
2. Bundesliga	18	18.00	Germany	2004	85.71
La Liga	18	20.00	Spain	2005	90.00
La Liga 2	16	22.00	Spain	2007	88.89
Liga Portugal 2	16	18.25	Portugal	2007	88.89
Série A (Brazil)	15	20.00	Brazil	2007	83.33
Challenger Pro League	14	15.43	Belgium	2006	73.68
Ligue 2	13	20.00	France	2004	61.90
J1 League	13	18.31	Japan	2009	81.25
J2 League	13	21.69	Japan	2011	92.86
Série B (Brazil)	13	20.00	Brazil	2010	86.67
Jupiler Pro League	13	16.31	Belgium	2008	76.47
Saudi Pro League	7	15.43	Saudi Arabia	2014	63.64
Süper Lig	5	19.00	Turkey	2014	45.45

The preprocessing pipeline culminated into two high-fidelity datasets, *Validated Standings* and *Validated Games*, which restrict the analytical scope exclusively to the seasons and championships verified as structurally sound. To ensure longitudinal consistency, we implemented an Entity Resolution protocol: raw strings for teams and leagues underwent sanitization—involving case normalization and the removal of diacritics—followed by a canonical mapping of team names to unique identifiers.

This procedure accounts for historical club rebrands and nomenclature variations, ensuring that the deterministic unique identifiers¹ remain invariant across the 306 valid seasons. Furthermore, temporal variables were subjected to rigorous parsing and conversion into unified *datetime* objects, supplemented by manual curation via targeted online research to resolve persistent null values. These consolidated dataframes provide the empirical foundation for the statistical assessment of tournament efficacy and schedule-driven discouragement effects explored in the subsequent sections.

4 Methodology

The analytical framework of this study isolates the impact of temporal schedule sequencing on tournament outcomes. We transform raw data into a structured inferential pipeline comprising three stages: (i) computation of the Strength of Schedule Balance,

¹ To guarantee data integrity and record traceability, we employed the SHA-256 (Secure Hash Algorithm 256-bit) to generate fixed-size, collision-resistant cryptographic hashes for each entry. This ensures that identifiers remain invariant across datasets regardless of string variations in raw sources.

(ii) heuristic-based classification of schedule types, and (iii) non-parametric statistical evaluation of ranking disparities conditioned on schedule type.

4.1 Measuring Strength of Schedule Balance (SSB)

To quantify how opponent difficulty is distributed across a team’s schedule, we introduce the *Strength of Schedule Balance (SSB)* metric. Let $R = \{r_1, \dots, r_n\}$ represent the set of teams ordered by a latent strength or skill rank, denoted as s_i , such that r_1 is the strongest team and r_n is the weakest. In our evaluation, this skill rank s_i is empirically measured using two proxies: the team’s final league standing (the oracle rank) and the team’s market value.

For a specific team r_i , the schedule S_i is the ordered sequence of opponents faced in the first half of the season. We define the Strength of Schedule Balance, $\rho(S_i)$, as the Spearman’s rank correlation coefficient between the actual schedule S_i of team r_i and a theoretical perfectly unbalanced schedule R_{-i} . In this generalized model, R_{-i} is the ordered list of all opponents excluding r_i sorted by their strength s_i . For example, in a tournament composed of four teams, $R_{-2} = \{r_1, r_3, r_4\}$ denotes the perfectly unbalanced schedule of the second-best team r_2 . In this formulation, $\rho(S_i) \in [-1, 1]$ serves as a proxy for the “loading” of the schedule:

- $\rho(S_i) \approx 1$: Indicates a **front-loaded** or **unbalanced-strong** schedule, in which the team faces elite opponents early in the competition, with difficulty decreasing over time.
- $\rho(S_i) \approx -1$: Indicates a **back-loaded** or **unbalanced-weak** schedule, in which the team initially faces weaker opponents, with difficulty escalating toward the season’s conclusion.
- $\rho(S_i) \approx 0$: Indicates a **balanced** schedule, characterized by a uniform distribution of opponent difficulty across rounds.

4.2 Threshold Selection and Group Classification

To facilitate comparative analysis, we categorize the continuous $\rho(S_i)$ values into three discrete cohorts: **Balanced** (G_0), **Unbalanced-Weak** (G_-), and **Unbalanced-Strong** (G_+). We consider two criteria for assigning schedules to these categories. First, by the magnitude of $\rho(S_i)$:

Definition 1 (Magnitude Criterion). *Given a threshold $\tau \in [0, 1]$, a schedule S_i is categorized according to the value of $\rho(S_i)$ as follows:*

- G_0 (**balanced**): $-\tau \leq \rho(S_i) \leq \tau$, corresponding to a balanced schedule.
- G_- (**unbalanced-weak**): $\rho(S_i) < -\tau$, corresponding to a back-loaded schedule.
- G_+ (**unbalanced-strong**): $\rho(S_i) > \tau$, corresponding to a front-loaded schedule.

Second, based on the statistical significance of $\rho(S_i)$, assessed via Spearman’s rank correlation coefficient between S_i and R_{-i} :

Definition 2 (Statistical Significance Criterion). Given a significance level $\alpha \in (0, 1)$ and the corresponding p -value obtained from Spearman’s rank correlation test between S_i and R_{-i} , a schedule S_i is categorized as follows:

- G_0 (**balanced**): $p \geq \alpha$, indicating no statistically significant schedule loading.
- G_- (**unbalanced-weak**): $p < \alpha$ and $\rho(S_i) < 0$, indicating a statistically significant back-loaded schedule.
- G_+ (**unbalanced-strong**): $p < \alpha$ and $\rho(S_i) > 0$, indicating a statistically significant front-loaded schedule.

The magnitude-based and statistical-significance criteria capture complementary aspects of schedule imbalance. The magnitude criterion offers a direct and interpretable measure of the extent to which a schedule is front- or back-loaded, enabling controlled comparisons across different threshold values. In contrast, the statistical-significance criterion accounts for uncertainty arising from schedule length and opponent variability, distinguishing meaningful temporal patterns from fluctuations that may occur by chance. Together, these criteria allow us to distinguish structurally unbalanced schedules from those that are only weakly or spuriously correlated with opponent strength, thereby supporting a more robust comparative analysis.

As expected, the resulting group distributions depend on the choice of parameters τ and α . To ensure the robustness of the **SSB** metric, we evaluated its behavior using two distinct proxies for team skill: final league standings and squad market valuations. Table 2 reports the proportion of schedules assigned to each group across these classification frameworks, computed over the validated dataset.

Table 2. Group distributions across different classification thresholds for ranking and market value classifications.

Classification Criterion	Final Standing			Market Value		
	G_- (%)	G_0 (%)	G_+ (%)	G_- (%)	G_0 (%)	G_+ (%)
$\alpha = 0.10$ (Significance)	4.1	91.8	4.1	4.5	91.9	3.7
$\alpha = 0.05$ (Significance)	1.9	96.1	2.0	2.3	96.1	1.6
$\tau = 0.2$ (Magnitude)	19.5	61.3	19.2	19.2	61.7	19.1
$\tau = 0.3$ (Magnitude)	9.6	80.9	9.5	9.5	81.5	9.0
$\tau = 0.4$ (Magnitude)	3.8	92.3	3.9	4.2	92.4	3.4

The empirical distribution reveals a high degree of convergence between the two proxies. While p -value thresholds ($\alpha = 0.05$) offer high statistical confidence, they result in extremely sparse groups ($< 2.5\%$) that lack the variance necessary to detect systemic bias. Conversely, $\tau = 0.2$ captures nearly 40% of the population as unbalanced, potentially diluting the effect of extreme scheduling.

Consequently, we adopted the **Magnitude Criterion** with $\tau = 0.3$ as the optimal threshold for both classification types. This heuristic ensures that approximately **10%** of the total sample is captured in each unbalanced extreme (G_- and G_+), providing a robust and balanced subset for testing the discouragement and momentum effect hypotheses. The alignment between ranking-based and valuation-based distributions sug-

gests that the structural loading of a schedule is a persistent feature, independent of how team quality is operationalized.

4.3 Empirical Illustrations of Schedule Imbalance

To illustrate the structural variance captured by the **SSB** metric, we examine the Premier League in the season 2024/2025 using both final league rankings and squad market valuations.

Consider Chelsea, who finished 4th in the final standings. When utilizing the final ranking as the strength proxy (s_i), their schedule $\mathbf{S}_4$ is represented as the sequence of opponent ranks: $\mathbf{S}_4 = \{3, 16, 12, 9, 14, 8, 7, 1, 5, 15, 2, 18, 6, 20, 17, 10, 13, 11, 19\}$. This yields a Strength of Schedule Balance of $\rho(\mathbf{S}_4) = 0.3474$. When applying the market value proxy—where Chelsea is ranked 2nd in squad valuation—the schedule sequence becomes $\mathbf{S}_2 = \{1, 13, 10, 11, 15, 9, 12, 4, 8, 5, 3, 20, 7, 18, 6, 14, 16, 17, 19\}$, resulting in a higher correlation of $\rho(\mathbf{S}_2) = 0.4351$. In both frameworks, Chelsea is categorized into the **unbalanced-strong** group (G_+), indicating a consistently identified front-loaded schedule.

In contrast, Leicester City (ranked 18th in standings and 20th in market value) provides an illustration of a back-loaded or **unbalanced-weak** schedule. Under the final ranking proxy, their schedule $\mathbf{S}_{18} = \{17, 11, 6, 12, 13, 2, 9, 20, 7, 19, 15, 4, 10, 14, 8, 5, 16, 1, 3\}$ yields $\rho(\mathbf{S}_{18}) = -0.3053$. Using the market value proxy, the sequence $\mathbf{S}_{20} = \{6, 17, 7, 10, 16, 3, 11, 18, 12, 19, 5, 2, 14, 15, 9, 8, 13, 4, 1\}$ results in $\rho(\mathbf{S}_{20}) = -0.2386$.

4.4 Inferential Framework: Efficacy and Effect Size

The terminal stage of our methodology assesses whether the group assignment (G_- , G_0 , G_+) significantly distorts the final league standings. Formally, let $R = \{r_1, r_2, \dots, r_n\}$ denote the ordered list of teams according to their final positions in the league standings, where r_1 is the champion, r_2 the runner-up, and so on. Let $G(r_i) \in \{G_-, G_0, G_+\}$ denote the group assignment of team r_i . We formulate our hypothesis as follows:

- H_0 : The final league ranking is independent of the group assignment,
 $\Pr(r_i \mid G(r_i) = G_-) = \Pr(r_i \mid G(r_i) = G_0) = \Pr(r_i \mid G(r_i) = G_+), \forall i.$
- H_1 : The group assignment distorts the final standings,
 $\exists i$ such that $\Pr(r_i \mid G(r_i) = g) \neq \Pr(r_i \mid G(r_i) = g'),$ for some $g \neq g'.$

Mann-Whitney U Test for Distributional Shift. We begin testing these hypotheses using the non-parametric Mann–Whitney U test to compare the distributions of final ranks between pairs of groups (e.g., G_+ vs. G_0). Because the test is rank-based and makes no assumptions about the normality of the data, it is well suited to ordinal outcomes such as league standings. The test assesses the null hypothesis that team rank is independent of group assignment. A statistically significant result ($p < 0.05$) indicates that certain ranks are disproportionately represented in one group relative to another,

suggesting that schedule imbalance induces a systematic advantage or disadvantage that the round-robin format does not fully neutralize.

Effect Size: Cliff’s Delta (δ). Because our large sample size ($N = 5,944$) may yield significant p-values even for negligible differences, we employ **Cliff’s Delta** (δ) to quantify the actual magnitude of the effect. This non-parametric effect size metric, bounded by $[-1, 1]$, measures how often values in one distribution are larger than values in another, providing a robust evaluation of distributional overlap.

Formally, we compare the rank of the teams in a unbalanced group $\mathcal{U} \in \{G_+, G_-\}$ with the rank of the teams in the balanced group $\mathcal{B} \equiv G_0$. For a given rank $u \in \mathcal{U}$ and rank $b \in \mathcal{B}$, we define the dominance function $d(u, b)$ as:

$$d(u, b) = \begin{cases} +1 & \text{if } u > b \quad (\text{Unbalanced performed worse}) \\ -1 & \text{if } u < b \quad (\text{Unbalanced performed better}) \\ 0 & \text{if } u = b \end{cases} \quad (1)$$

We apply this framework in two distinct contexts to isolate the impact of strength of schedule imbalance:

1. Season-Level Effect Size (δ). To analyze individual seasons, we compute the standard Cliff’s Delta as a measure of effect size. Let $\mathcal{U}$ and $\mathcal{B}$ denote the sets of final ranks for teams with unbalanced and balanced schedules in a given season, with cardinalities $n_{\mathcal{U}}$ and $n_{\mathcal{B}}$, respectively. The seasonal effect size is defined as the average dominance across all pairwise comparisons between ranks in $\mathcal{U}$ and $\mathcal{B}$:

$$\delta = \frac{1}{n_{\mathcal{U}}n_{\mathcal{B}}} \sum_{u \in \mathcal{U}} \sum_{b \in \mathcal{B}} d(u, b) \quad (2)$$

A positive δ indicates that teams with unbalanced schedules tend to finish in worse positions (i.e., higher rank values) than those with balanced schedules. For instance, if every rank in $\mathcal{B}$ is strictly lower than every rank in $\mathcal{U}$, then $\delta = 1$, reflecting complete dominance of balanced over unbalanced teams.

2. League-Wide Stratified Effect Size ($\delta_{\mathcal{L}}$). To aggregate results across an entire league history while controlling for inter-season and inter-league variance, we implement a **League-Wide Stratified Cliff’s Delta**. This ensures that a team is only compared against its contemporaries within the same season, preventing confounding factors — such as varying league sizes or historical shifts in competitive balance — from skewing the results. For a league $\mathcal{L}$ comprising m seasons $L_1, L_2, \dots, L_m$, we compute the aggregated effect size as

$$\delta_{\mathcal{L}} = \frac{\sum_{L \in \mathcal{L}} \sum_{u \in \mathcal{U}_L} \sum_{b \in \mathcal{B}_L} d(u, b)}{\sum_{L \in \mathcal{L}} (|\mathcal{U}_L| \times |\mathcal{B}_L|)}, \quad (3)$$

where $\mathcal{B}_L$ and $\mathcal{U}_L$ denote the sets of balanced and unbalanced teams in season L , with cardinalities $|\mathcal{B}_L|$ and $|\mathcal{U}_L|$, respectively. This formulation corresponds to a weighted

average of the seasonal Cliff’s Delta values, where each season is weighted by the total number of pairwise comparisons between balanced and unbalanced teams.

This stratified approach allows for a clean isolation of the schedule’s structural impact across our longitudinal dataset, summarizing whether specific competitions consistently penalize unbalanced schedules over decades of play without confounding the data across different years.

Finally, to interpret the magnitude of the resulting coefficients, we adopt the thresholds established by Romano et al. [27]. Specifically, an effect size is considered **negligible** if $|\delta| < 0.147$, **small** if $|\delta| < 0.33$, **medium** if $|\delta| < 0.474$, and **large** otherwise.

5 Results

5.1 Prevalence of Schedule Imbalance

Table 3 summarizes the schedule classifications across the 20 competitions by highlighting the most extreme deviations alongside the global average. Both metrics reveal high structural consistency, with the English Championship consistently demonstrating the highest stability ($G_0 > 85\%$). Additionally, Brasileiro Série A exhibits the highest financial balance ($G_0 = 89.00\%$). Conversely, the Saudi Pro League is the most volatile under the ranking proxy ($G_0 = 68.52\%$), while Liga Portugal 2 shows a significant divergence, dropping to a 68.49% balance rate under the market value classification. Despite these localized outliers, the global average for balanced schedules (G_0) securely anchors at approximately 80% across both proxies, confirming the double round-robin format’s overall structural resilience.

5.2 Aggregate Impact on Ranking Efficacy

In this section, we examine whether the observable imbalance identified in the previous section translates into distortions in final league standings. To this end, we first analyze the aggregate distribution of final positions across the dataset.

To quantify the magnitude of these distributional shifts, we analyze the effect sizes for individual seasons. Figure 1 presents the distribution of Cliff’s Delta (δ) values for the G_- vs. G_0 and G_+ vs. G_0 comparisons. The results indicate a conditional rather than universal effect. Under the traditional final ranking classification, the median effect size for both groups is centered strictly at zero ($\delta \approx 0.00$), but the inter-quartile ranges (IQRs) reveal a distinct directional asymmetry. The G_- cohort (Easy Start) consistently extends into the negative “medium” effect zone ($-0.474 < \delta < -0.33$), suggesting that while an easy start does not guarantee success, it significantly widens the potential for overperformance. Conversely, the G_+ cohort (Hard Start) exhibits a mirrored skew toward the positive “medium” zone, indicating a higher susceptibility to underperformance.

Crucially, this directional asymmetry persists when transitioning to the exogenous market value proxy. While substituting the final standing with pre-season financial data shifts the median of the G_- cohort slightly below zero, the overall spread and structural behavior remain consistent across both methodologies. This comparison shows a consolidation of this variance into the directional patterns described above, supporting the

Table 3. Prevalence of schedule types across the 20 analyzed football leagues. For each league, the table reports the percentage of schedules classified as unbalanced-weak (G_-), balanced (G_0), and unbalanced-strong (G_+) when **SSB** is computed using two alternative team-strength proxies: final standing and market value. The column N denotes the number of valid schedules available for that league, and the last row reports the global average across all leagues.

League Name	N	Final Standing			Market Value		
		% G_-	% G_0	% G_+	% G_-	% G_0	% G_+
Saudi Pro League	108	16.67	68.52	14.81	12.96	74.07	12.96
Challenger Pro League	216	12.96	72.69	14.35	11.11	80.56	8.33
Bundesliga 2	324	13.27	75.93	10.80	9.57	77.47	12.96
Jupiler Pro League	212	13.21	76.42	10.38	11.32	79.72	8.96
J1 League	238	12.61	77.31	10.08	12.18	76.89	10.92
Bundesliga	324	9.57	79.01	11.42	7.10	83.95	8.95
Brasileiro Série B	260	10.38	79.62	10.00	9.23	80.00	10.77
Liga Portugal 2	292	8.90	79.79	11.30	15.41	68.49	16.10
Ligue 2	260	10.00	80.77	9.23	11.15	78.85	10.00
Premier League	380	10.00	80.79	9.21	10.00	81.58	8.42
Liga Portugal	322	9.32	81.37	9.32	8.70	81.68	9.63
Serie A	380	10.79	81.58	7.63	8.42	82.89	8.68
La Liga 2	352	7.95	82.10	9.94	10.80	78.98	10.23
J2 League	282	8.16	82.27	9.57	7.45	82.98	9.57
Serie B	407	8.35	82.56	9.09	9.83	81.82	8.35
Ligue 1	376	8.78	82.71	8.51	8.78	84.84	6.38
La Liga	360	6.94	83.61	9.44	9.44	83.06	7.50
Brasileiro Série A	300	7.33	84.33	8.33	6.00	89.00	5.00
Süper Lig	95	10.53	85.26	4.21	8.42	87.37	4.21
Championship	456	7.46	85.96	6.58	7.46	86.62	5.92
Global Average (20 Leagues)	5944	10.16	80.13	9.71	9.77	81.04	9.19

hypothesis that schedule imbalances act as a latent risk factor rather than a deterministic outcome.

5.3 League-Specific Significance

To ensure that the aggregate patterns are not masking localized anomalies, we analyze the impact of schedule imbalance on a per-league basis using the Stratified Cliff’s Delta ($\delta_{\mathcal{L}}$) and Mann–Whitney significance tests. We perform this analysis using the **SSB** metric under both market value and final standing proxies, which yield broadly similar qualitative patterns. However, we present the plots using the market value proxy in order to reduce concerns regarding endogeneity. Figure 2 maps each league according to its effect size (y-axis) and statistical significance (x-axis, $-\log_{10}(p\text{-value})$), while Table 4 compares the significant results between proxy choices as a robustness check.

Overall, both proxies suggest that the double round-robin format is broadly robust: most leagues remain to the left of the $p = 0.05$ threshold, and the major European leagues do not exhibit statistically significant deviations under either schedule extreme.

The clearest league-specific effects emerge for unbalanced-weak schedules (G_- vs. G_0). In this comparison, the Jupiler Pro League ($\delta_{\mathcal{L}} = -0.643$, $p < 10^{-5}$) and Liga

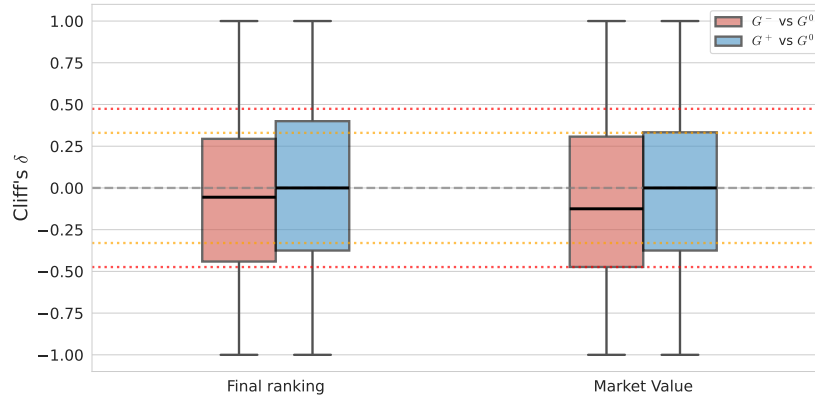


Fig. 1. Cliff's δ distributions for ranking efficiency comparing evaluation metrics. Comparisons are shown for G_- vs. G_0 (red) and G_+ vs. G_0 (blue) using both Final Ranking and Market Value to compute the Strength of Schedule Balance. Dotted lines denote effect size thresholds: small ($|\delta| = 0.33$, yellow) and medium ($|\delta| = 0.474$, red). The grey dashed line at $\delta = 0$ represents the null effect.

Portugal 2 ($\delta_{\mathcal{L}} = -0.562$, $p < 10^{-6}$) stand out as the strongest anomalies, with additional significant negative effects in the Saudi Pro League ($\delta_{\mathcal{L}} = -0.356$, $p = 0.023$), Ligue 2 ($\delta_{\mathcal{L}} = -0.278$, $p = 0.024$), and Campeonato Brasileiro Série B ($\delta_{\mathcal{L}} = -0.265$, $p = 0.038$). As shown in Table 4, the Jupiler Pro League and Saudi Pro League remain significant under both proxy choices. Because negative Cliff's Delta indicates that teams in the unbalanced group tend to finish in better positions than those in the balanced baseline, these results are consistent with a momentum-type advantage associated with easier starts.

In contrast, the evidence for unbalanced-strong schedules (G_+ vs. G_0) is more limited and does not support a generalized discouragement pattern under the market value proxy. No league presents a statistically significant positive $\delta_{\mathcal{L}}$, and the only significant deviations are observed in J2 League ($\delta_{\mathcal{L}} = -0.254$, $p = 0.040$) and Campeonato Brasileiro Série B ($\delta_{\mathcal{L}} = -0.254$, $p = 0.034$), both indicating that teams with front-loaded difficult schedules finished better than their balanced counterparts. Unlike the G_- comparison, these league-level results do not align closely with the findings obtained under the final standing proxy, which suggests that the evidence for hard-start effects is less stable across alternative definitions of team strength.

Taken together, these findings indicate that league-level anomalies are concentrated in a small subset of competitions and are not symmetric across schedule extremes. Thus, the evidence is more consistent with league-specific associations than with a universal schedule mechanism affecting tournament efficacy.

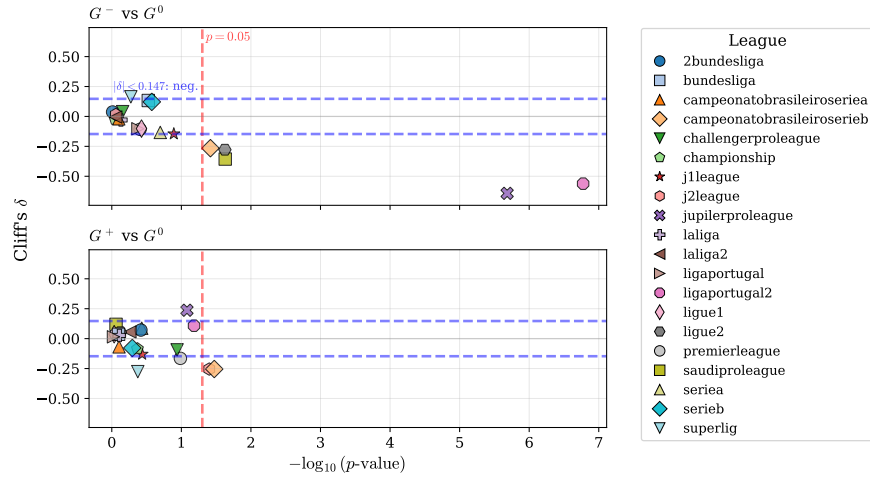


Fig. 2. Significance tests per league for the G_- and G_+ groups based on market value classification. Points to the right of the red dashed line ($p < 0.05$) represent statistically significant deviations from the balanced baseline. The region between the blue dashed lines ($|\delta| \leq 0.147$) denotes a negligible effect size.

6 Discussion

The main contribution of this study is the empirical evaluation of whether the temporal ordering of opponent strength affects tournament efficacy in double round-robin football leagues. At the aggregate level, the evidence points to a predominantly robust format: most competitions do not exhibit large or systematic distortions in final rank distributions attributable solely to schedule sequencing.

At the same time, the results do not support a purely neutral view of fixture ordering. The distributional analysis suggests that schedule imbalance acts less as a deterministic force and more as a conditional source of variation, with some seasons exhibiting noticeable deviations even when the median effect remains close to zero. In this sense, the timing of strong and weak opponents appears capable of shaping competitive trajectories in specific settings, even if it does not consistently alter outcomes across the ecosystem as a whole.

The clearest league-level deviations emerge for unbalanced-weak schedules (G_-), where several competitions display significant negative effect sizes relative to the balanced baseline. Because negative Cliff's Delta indicates that teams in the unbalanced group tend to finish in better final positions, these patterns are consistent with an advantage associated with easier starts. The recurrence of this signal across multiple leagues, and its partial stability across alternative strength proxies, suggests that back-loaded schedules may in some environments create a meaningful pathway to overperformance.

By contrast, the evidence for unbalanced-strong schedules (G_+) is considerably weaker. Although isolated league-level deviations appear under both strength definitions, they are fewer in number, less stable across proxies, and do not reveal a coherent

Table 4. Comparison of league-level results that are statistically significant in the Mann–Whitney tests (p -value < 0.05) under the two **SSB** proxy choices. Results are reported separately for the G_- vs. G_0 and G_+ vs. G_0 comparisons.

Proxy	Sig. Better ($\delta_{\mathcal{L}} < 0$)	Sig. Worse ($\delta_{\mathcal{L}} > 0$)
G_- vs. G_0		
Final Standing	Saudi Pro; Jupiler Pro	—
Market Value	Saudi Pro; Jupiler Pro; Liga Portugal 2; Ligue 2; Camp. Bras. Série B	—
G_+ vs. G_0		
Final Standing	J1 League; Liga Portugal 2	Jupiler Pro
Market Value	J2 League; Camp. Bras. Série B	—

Notes: “Sig. Better” denotes significantly better final ranks for the unbalanced group (negative $\delta_{\mathcal{L}}$); “Sig. Worse” denotes significantly worse ranks (positive $\delta_{\mathcal{L}}$). “—” indicates no leagues in that category for the given proxy.

pattern of systematic underperformance. As a result, the present findings provide limited support for a generalized discouragement effect linked to front-loaded schedules.

This asymmetry between G_- and G_+ is substantively important. If schedule imbalance operated through a uniform mechanism, one would expect easier starts and harder starts to generate roughly symmetric effects in opposite directions. Instead, the empirical picture is more uneven: easier starts appear more consistently associated with advantageous outcomes than harder starts are with disadvantageous ones. This suggests that any schedule effect is likely mediated by contextual factors such as league stratification, competitive volatility, or the interaction between early results and the broader dynamics of a season.

The comparison between market value and final standing proxies also helps clarify the interpretation of the **SSB** metric. On the one hand, the persistence of the broad qualitative pattern under both definitions indicates that the metric is not merely capturing an artifact of a single operationalization of team strength. Although some league-specific anomalies differ across proxy choices, the overall empirical pattern remains consistent.

From a tournament-design perspective, the results suggest that schedule sequencing is usually not the dominant driver of ranking efficacy, particularly in competitions where differences in team quality are strong enough to be revealed over a full double round-robin season. However, the existence of recurrent anomalies in a subset of leagues indicates that fixture ordering should not be treated as a purely logistical detail. Even under a formally symmetric format, the schedule may still influence the path by which teams accumulate points and, in some environments, affect the degree to which final standings reflect underlying team strength and thus tournament efficacy.

7 Conclusion

This paper introduced the Strength of Schedule Balance (**SSB**) metric and applied a stratified non-parametric framework to evaluate whether the temporal ordering of opponent strength affects tournament efficacy in professional football leagues. Using a validated longitudinal dataset of 306 seasons across 20 competitions, we examined whether balanced, unbalanced-weak, and unbalanced-strong schedules are associated with systematic differences in final league standings.

Our findings indicate that the double round-robin format is generally robust to schedule imbalance. At the aggregate level, most competitions do not exhibit large or systematic distortions in final ranking outcomes attributable solely to fixture sequencing. However, league-level analysis reveals localized anomalies, particularly for unbalanced-weak schedules, where easier starts are more consistently associated with advantageous outcomes than harder starts are with disadvantageous ones. Overall, the evidence provides limited support for a generalized discouragement effect under front-loaded difficult schedules, while suggesting that back-loaded schedules may, in some competitive environments, be associated with overperformance.

8 Limitations and Future Work

This study has important limitations. First, although the **SSB** metric is defined as a continuous measure of schedule imbalance, our empirical analysis relies on stratifying it into discrete groups. This design facilitates interpretation and non-parametric comparison, but it may obscure more gradual relationships between schedule imbalance and tournament efficacy.

A natural extension would be to analyze the impact of **SSB** as a continuous variable. In principle, this could be done through a regression-based framework that estimates the coefficient associated with schedule imbalance and its statistical significance. However, such an approach raises methodological challenges, including the choice of an appropriate model specification, the treatment of potential non-linearity, the dependence structure among covariates, and the need to control for confounding factors that may also influence final standings, such as market value, injuries, and home-away scheduling effects.

Future work should address these issues and explore whether schedule imbalance affects outcomes beyond final rankings. In particular, it would be valuable to investigate its relationship with other indicators of competitive dynamics, such as stadium attendance or other measures of team performance and engagement.

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9 Appendix: Sensitivity Analysis of Threshold Parameters

To assess whether the league-specific effects identified in the main analysis depend on the choice of classification threshold, we performed a sensitivity analysis across alternative magnitude criteria, $\tau \in \{0.2, 0.3, 0.4\}$, as well as a statistical-significance criterion ($\alpha = 0.1$) based on the Spearman correlation test used in the definition of the SSB metric $\rho(\mathbf{S})$. The results, summarized in Table 5, indicate that the broad qualitative conclusions of the paper are robust to parameter variation, although the stability of league-level anomalies differs markedly across schedule extremes and proxy choices.

Stability in G_- vs. G_0 . The most persistent pattern appears in the comparison between unbalanced-weak and balanced schedules. Under the final-standing proxy, the Saudi Pro League and Jupiler Pro League remain in the statistically significant “better” zone across all magnitude thresholds, while under the market-value proxy the Jupiler Pro League and Liga Portugal 2 recur consistently, with the Saudi Pro League also appearing at $\tau = 0.3$ and $\tau = 0.4$. This repeated concentration of significant negative effects supports the interpretation that the main league-specific anomalies are more stable for easier starts than for harder starts.

Instability in G_+ vs. G_0 . By contrast, the results for unbalanced-strong schedules are considerably less stable. Under the final-standing proxy, the significant leagues vary across thresholds, with Jupiler Pro League and La Liga appearing in the “worse” zone at $\tau = 0.2$, while J1 League and J2 League appear repeatedly in the “better” zone and

Table 5. Leagues in statistically significant zones across thresholds τ and α , by proxy

Scenario	Thresh.	<i>Final Standing Proxy</i>		<i>Market Value Proxy</i>	
		Sig. worse	Sig. better	Sig. worse	Sig. better
G_- vs. G_0	$\tau = 0.2$	—	Süper Lig; C. Br. B; Saudi Pro; Jupiler Pro; Champ.	—	Ligue 2; Liga Portugal 2; Jupiler Pro
	$\tau = 0.3$	—	Saudi Pro; Jupiler Pro	—	C. Br. B; Ligue 2; Saudi Pro.; Jupiler Pro.; Lig. Port. 2
	$\tau = 0.4$	—	Saudi Pro; Jupiler Pro	—	Serie A; Saudi Pro.; Jupiler Pro.; Liga Portugal 2
	$\alpha = 0.1$	—	Jupiler Pro; Ligue 2	—	Serie A; Jupiler Pro; Lig. Port. 2
G_+ vs. G_0	$\tau = 0.2$	Jupiler Pro; La Liga	J1 League; J2 League	Jupiler Pro	J2 League
	$\tau = 0.3$	Jupiler Pro	J1 League; J2 League	—	J2 League; C. Br. B
	$\tau = 0.4$	—	C. Brasileiro A; J1 League	—	—
	$\alpha = 0.1$	—	C. Brasileiro A; J1 League	Jupiler Pro; Lig. Port. 2	—

Notes: “—” indicates no leagues in that category.

Campeonato Brasileiro Série A emerges only at stricter definitions. Under the market-value proxy, the pattern is even weaker: J2 League is the only league recurring in the “better” zone for $\tau = 0.2$ and $\tau = 0.3$, while no significant leagues remain at $\tau = 0.4$. Overall, these results do not indicate a robust or universal discouragement effect.

Role of the significance-based criterion. The significance-based grouping ($\alpha = 0.1$) yields conclusions that are directionally consistent with the threshold analysis, but it also highlights the sensitivity of specific leagues to the grouping rule. In the G_- comparison, Jupiler Pro League remains significant under both proxies, while Ligue 2, Serie A, and Liga Portugal 2 appear depending on the choice of team-strength proxy. In the G_+ comparison, the significant leagues differ substantially across proxies, reinforcing the view that hard-start effects are less stable and more dependent on the operational definition of imbalance.

Justification for $\tau = 0.3$. These findings support the use of the Magnitude Criterion with $\tau = 0.3$ in the main analysis. Relative to $\tau = 0.2$, this threshold reduces

the inclusion of weaker or noisier cases while preserving the recurring league-specific anomalies. Relative to $\tau = 0.4$, it avoids an overly restrictive definition that removes a substantial portion of the detectable signal. Thus, $\tau = 0.3$ provides a reasonable balance between sample size and effect isolation in the identification of schedule-related distortions.